Type MS · Interpret human growth data, proportions and percentile charts

Answer 1 of 4 · Question 8 part c.ii · 3 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part c.ii in this mark scheme extract

Number					
8	(a)	(i)	triggers ovulation/causes release of the egg;		1
		(ii)	Three from: • FSH stimulates growth/production of follicles in the ovary; • more eggs mature/produced; • increases the chances of more than one egg being released; • each egg is fertilised by a sperm;		3
	(b)	(i)	Three from: • placenta supplies glucose/oxygen from the mother; • twins have to share the supply of glucose/receive less glucose/oxygen; • less aerobic respiration; • less energy is released in the foetus and less growth occurs;	Allow nutrients for mp's 1 and 2	3
		(ii)	Two from • same egg and sperm used; • zygote splits/divides • forming 2 embryos;		2
	(c)	(i)	75th		1
		(ii)	• baby is weighed regularly; • growth should follow a percentile/graph/compare with (growth) chart/graph/compare mass to others of the same age; • detect if the baby gains or loses too much mass;		3

Total 13 marks

Type MS · Interpret human growth data, proportions and percentile charts

Answer 1 of 4 · Question 8 part c.ii · 3 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part c.ii in this mark scheme extract

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Type MS · Interpret human growth data, proportions and percentile charts

Answer 2 of 4 · Question 5 part a.ii · 2 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

5	(a)	(i)	175.9 + 178.5 + 166.3 + 167.6 ÷ 4;		1
		Clip with table	172.1;	Full marks for correct final answer Reject 172	1
		(ii)	Height increases with age (for males / females / both); Males are taller/females are shorter (at all ages); Females increase in height greater compared with males between the ages of 12 and 18; Greatest height increase between ages 12 to 14;		Max 2
		(iii)			1
			Use more students in each age group; Use a greater age range of students; Use ages with 1 year gap/13,15,17;	Allow other valid improvements Ignore repeats	1
	(b)	(i)	71 ÷ 1.46^2;		1
		Clip with table	33/33.3;	Full marks for final correct answer One mark for correct calculation with incorrect figures	1
		(ii)	student Y is obese;	ECF from b(i)	1
		CLIP (b)(i) with (b)(ii)	higher risk of diabetes/heart disease/arthritis/cancers;	Allow other medical condition linked to obesity	1

Total question 5 = 10 marks

Type MS · Interpret human growth data, proportions and percentile charts

Answer 2 of 4 · Question 5 part a.ii · 2 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Interpret human growth data, proportions and percentile charts

Answer 3 of 4 · Question 4 part b · 2 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

4(a)(i)	¼/25%/1:4/0.25;		**(1)**

Question Number	Answer	Additional guidance	Mark
4(a)(ii)	½/50%/1:2/0.5/2 4;		**(1)**

Question Number	Answer	Additional guidance	Mark
4(b)	head decreases (in proportion); legs increase (in proportion);		**(2)**

Question Number	Answer	Additionalg uidance	Mark
4(c)	• allowance for babies of different height; • allows comparison to norm (in population); • determine changes in proportion; • over time / indicates/shows which parts of body are developing/growing;		**Max (2)**

Total 6 Marks

Type MS · Interpret human growth data, proportions and percentile charts

Answer 3 of 4 · Question 4 part b · 2 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Interpret human growth data, proportions and percentile charts

Answer 4 of 4 · Question 4 part c · 2 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

4(a)(i)	¼/25%/1:4/0.25;		**(1)**

Question Number	Answer	Additional guidance	Mark
4(a)(ii)	½/50%/1:2/0.5/2 4;		**(1)**

Question Number	Answer	Additional guidance	Mark
4(b)	head decreases (in proportion); legs increase (in proportion);		**(2)**

Question Number	Answer	Additionalguidance	Mark
4(c)	• allowance for babies of different height; • allows comparison to norm (in population); • determine changes in proportion; • over time / indicates/shows which parts of body are developing/growing;		**Max (2)**

Total 6 Marks

Type MS · Interpret human growth data, proportions and percentile charts

Answer 4 of 4 · Question 4 part c · 2 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark